

# B5-BEYOND-LAYER-BOUND claim-level synthesis: the beyond-layer machinery (DR-2, STEP-5B, SC-SCOPE)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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## 1. Purpose and scope

Claim-level synthesis for B5-BEYOND-LAYER-BOUND. It adds no mathematics; every statement carries the unit's registered tier. **Scope:** the composition of B5's units into the canonical claim state, and the precise tier relations between the T5 claim and its T6/T7 units. **Not claimed:** the vacuum selection itself (that is B1/B2's head), arbitrary non-lattice DR-2, or full STEP-5B exhaustiveness beyond the amended-class pin.

## 2. The claim and its canonical state

**Claim (canonical).** Pattern-generic Gershgorin reduction of the beyond-layer bound: within the H-ADM-COH amended competitor class at second-cumulant order, the beyond-layer off-diagonal correction is bounded below the layer margin (T5 PINNED-CLOSURE; pinned by the operator decision record). No hypotheses; no open gates.

## 3. Sub-proof unit map (per-unit tier and role)

| Unit         | Tier                  | What it contributes                                                                                                                | Publication state                                         |
|--------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| DR-2/        | T7 (lattice, std. NT) | Lemma 1 backing: $E_+(Q) \leq_\epsilon R^\epsilon N^2$ decoupling-free on lattice shells; Lemma NT pinned in-bundle                | PUBLISHED – DR2-Lattice-T7-NTstandard-260612 (8/8, 38/38) |
| STEP-5B/     | T6                    | the rectangle prefactor $K(n) \leq 8 + 4\sqrt{14}\kappa^4\sqrt{n}$ + official threshold $n_{\text{adm}} \lesssim 1.59 \times 10^5$ | PUBLISHED – STEP-5B-Rectangle-T6-260612 (192/192)         |
| SC-SCOPE/    | T5 thin               | the (S) selection floor at the endpoint: joint = $\times 1.040 - \times 1.082 > 1$ on $\mathcal{W}_{\text{SC}}$                    | PUBLISHED – SC-SCOPE-T5-260612 (4/4, 29/29)               |
| H-LAYER-AUX/ | T5                    | RES-4 layer-ratio positivity $\rho(I) \geq \times 2.58 > 1$ , monotone (201 nodes) – input to STEP-5B's K-budget                   | AUXILIARY (operator-accepted v1.1)                        |
| T5-DOSSIER/  | T5                    | the assignment dossier pinning the claim tier (closure pillars, 192/192)                                                           | internal consolidation                                    |

## 4. Composition: why the claim is T5 while its units reach T7

The claim's statement is the AMENDED-CLASS closure at second-cumulant order – an operator-pinned decision-layer statement (H-ADM-COH adoption). Its strongest units exceed it: DR-2 is an unconditional lattice-class theorem (independent mathematical interest), STEP-5B's rectangle constant is theorem-grade T6, and SC-SCOPE certifies the endpoint floor at T5 thin. The claim tier is NOT the maximum of its units: it is pinned at the level of the weakest statement the CLAIM ITSELF makes (the class-amended, second-cumulant closure), per the T5-DOSSIER. Upgrading the claim would require the exhaustiveness decision layer (T-031: H-ADM-COH adoption record finalisation + the R-U6-1 backlog lemma, T-004),

not new analysis – the budget COMPARISON is already machine-closed ( $\times 55.6 / \times 8.8 / \times 2.1$ – $2.6$  across the intensity band, J-refined endpoint  $\times 2.6$ ).

Consumption map (who uses what): B1/B2’s head uses Lemma 1 (DR-2 mechanism) + Lemma 2 (in-head packing) for (O), SC-SCOPE for (S), STEP-5B for the K-budget; H-LAYER-AUX feeds STEP-5B. Per-edge accounting: parent capstone Main-Line-Synthesis-T013-260612 Sec. 3–4.

## 5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- ( $\alpha$ ) “A T5 claim holding a T7 unit invites tier-shopping: citing the claim at its unit’s tier.” ADRESSED: Sec. 4 fixes the rule – cite the CLAIM at T5 (its own statement), cite the UNIT at its bundle tier (its own statement). The two statements differ; the synthesis exists precisely to prevent the conflation.
- ( $\beta$ ) “SC-SCOPE thinness ( $\times 1.04$ ) could silently cap the whole B5 machinery.” VALID with mitigation: the thinness affects only the (S) endpoint; the off-diagonal chain (DR-2 + STEP-5B + RES-4) carries  $\geq \times 2.1$  margins everywhere on the band, and the head package consumes (S) with the thinness explicitly flagged (parent capstone Sec. 4a).
- ( $\gamma$ ) “The amended-class pin (H-ADM-COH) is an operator decision – is the claim then ‘proved’?” VALID and intrinsic: T5 PINNED-CLOSURE is the honest label for exactly this situation (closure WITHIN an operator-pinned class definition); the exhaustiveness question is tracked as T-031, not hidden.

**Quantitative sanity check (mandatory).** Cross-unit consistency: the STEP-5B rectangle bound at the class size ( $K(n_{\text{pack}}) \leq 113$ ) vs the measured  $K = 107$  (6%, safe side); RES-4’s  $\rho \geq 2.58$  vs the sharpened floor  $\rho_{\text{lat}} = 6.55$  (consistent: RES-4 predates the  $K \leq T'$  sharpening); SC-SCOPE joint  $\times 1.040$  reproduced from its terms  $[(1 + 0.385)/6.55 + (1 - 1/6.55)/1.13]^{-1}$ ; all three bundle digests verified at publication.

## 6. Result footer

|                         |                                                                                                                                                                                                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Result ID:              | B5-BEYOND-LAYER-BOUND-Synthesis-T013-260612<br>(PUBLISHED claim-level SYNTHESIS, T-013)                                                                                                                                                                                                                                                 |
| Precise statement:      | B5’s canonical state (T5 PINNED-CLOSURE, CLOSED@H-ADM-COH-AMENDED-CLASS, gates none) is the claim’s own decision-layer statement; its sub-proof units hold PUBLISHED bundles at their own tiers (DR-2 T7-lattice, STEP-5B T6, SC-SCOPE T5 thin) and are cited at those tiers by the B1/B2 head. Claim tier != max unit tier, by design. |
| Scope:                  | synthesis layer; no new math, no tier change, no gate flip.                                                                                                                                                                                                                                                                             |
| Evidence grade:         | SYNTHESIS over three PUBLISHED bundles + operator-accepted auxiliaries.                                                                                                                                                                                                                                                                 |
| Reproduction command:   | run the three bundles’ entry manifests (8 + 1 + 4 scripts).                                                                                                                                                                                                                                                                             |
| Falsifier:              | any falsifier of the cited footers; or a unit-tier mismatch against INDEX.md.                                                                                                                                                                                                                                                           |
| Tier before / after:    | no change (B5 stays T5 PINNED-CLOSURE).                                                                                                                                                                                                                                                                                                 |
| No-overclaim statement: | the claim is cited at T5; unit tiers are not claim tiers; exhaustiveness (T-031) and arbitrary-Q (T-030) remain open tracks.                                                                                                                                                                                                            |
| Next required action:   | T-031 decision-layer support (R-U6-1 lemma T-004 first); T-030 as research frontier.                                                                                                                                                                                                                                                    |